

Experiment No. 5

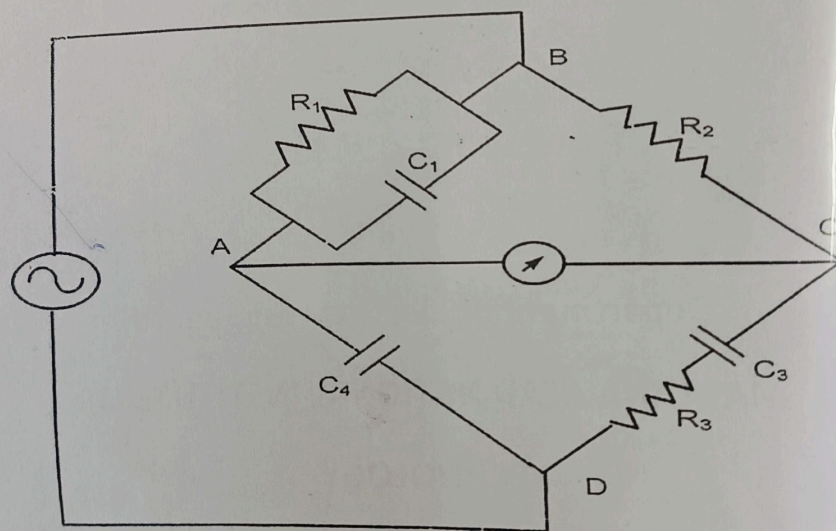
Aim: To determine the unknown Capacitance using Schering bridge method.

Apparatus:

1. Schering bridge Trainer
2. Patch cords
3. Multimeter

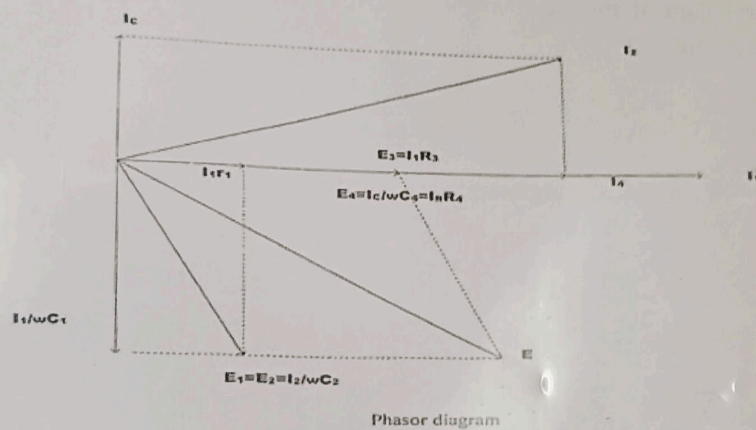
Theory:

The Schering bridge is used for measuring an unknown electrical capacitance and its dissipation factor. The dissipation factor of a capacitor is the ratio of its resistance to its capacitive reactance. The Schering bridge is basically a four-arm alternating current (AC) bridge circuit whose measurement depends on balancing the loads on its arms.



Schering Bridge

In the above shown Schering Bridge, the resistance R_1 and R_2 are known, while the resistance value of resistor R_3 is unknown. The capacitance values of C_1 and C_2 are also known, while the capacitance of C_3 is the value being measured. To measure R_3 and C_3 , the values of C_2 and R_2 are fixed, while the points A and B becomes zero. This happens when the voltages at points A and B are equal. In this case the bridge is said to be balanced.



When the bridge is balanced,

$$\frac{Z_1}{C_2} = \frac{R_2}{Z_3}$$

Where,

Z_1 is the impedance of R_1 in parallel with C_1

$$Z_1 = R_1 \left[2\pi f C_1 \left(\frac{1}{2\pi f C_1} + R_1 \right) \right]$$

$$Z_1 = R_1 (1 + 2\pi f C_1 R_1)$$

Z_3 is the impedance of R_3 in series with C_3

$$Z_3 = \frac{1}{2\pi f C_3} + R_3$$

The capacitors present in AC circuit contribute capacitive reactance to the impedance.

As the bridge is balanced, the negative and positive reactive components are equal and cancel out.

$$R_3 = C_1 * \frac{R_2}{C_2}$$

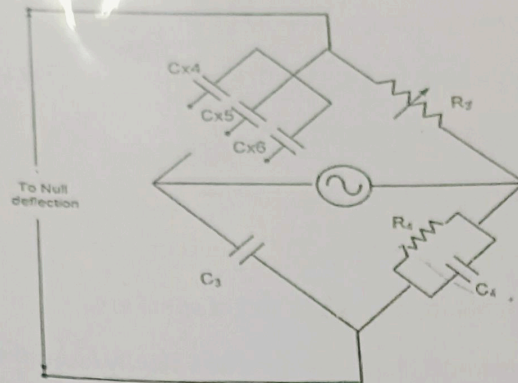
Similarly, the purely resistive components are equal.

$$\frac{C_2}{C_3} = \frac{R_2}{R_1}$$

$$C_3 = R_1 * \frac{C_2}{R_2}$$

Thus, from above equation, the unknown capacitance placed in the arm can be calculated for the following figure as

$$C_x = R_4 * \frac{C_3}{R_3}$$



Note that the balancing of a Schering bridge is independent of frequency. Its application is measuring of dielectric of insulation material.

Procedure:

1. Connect mains to the Trainer.
2. Connect terminal 15 to 12 (for evaluating unknown capacitor Cx_4).
3. Rotate variable Resistance R_3 towards anticlockwise direction.
4. Connect null detector (terminal 9 to 11 and 10 to 18).
5. Keep toggle of null detector towards 'off' condition.
6. Select frequency selector for any desired range of frequency.
 - 500 Hz to 1 kHz
 - 1 k Hz to 10 kHz
 - 10 kHz to 60 kHz
7. For example 2 kHz frequency, select frequency selector between the ranges 1 kHz-10kHz.

Observation Table:

S. No	Unknown capacitor	Frequency		Resistance $R_3 (\Omega)$	Resistance $R_4 (\Omega)$	Capacitance $C_3 (\mu F)$	C4
1.	Cx_4	f1					
		f2					
		f3					
2.	Cx_5	f1					
		f2					
		f3					
3.	Cx_6	f1					
		f2					
		f3					

Calculations

1. For unknown Capacitance Cx_4 on frequency f_1 :

$$Cx_4 = R_4 * \frac{C_3}{R_3}$$

$$= \text{_____} \mu F$$

Similarly calculate capacitance Cx_4 on frequency f_2 and f_3 and take the mean value.

2. For unknown Capacitance Cx_5 on frequency f_1 :

$$Cx_5 = R_4 * \frac{C_3}{R_3}$$

$$= \text{_____} \mu F$$

Similarly calculate capacitance Cx_5 on frequency f_2 and f_3 and take the mean value.

3. For unknown Capacitance Cx_6 on frequency f_1 :

$$Cx_6 = R_4 * \frac{C_3}{R_3}$$

$$= \text{_____} \mu F$$

Similarly calculate capacitance Cx_6 on frequency f_2 and f_3 and take the mean value.

Precautions:

1. Connect the circuit diagram with proper connection and take the measurements without error.
2. Calculate the value of capacitor without any error

Result:

The capacitance of given capacitor is _____ μF

Questions :

1. A Schering bridge can be used for the measurement of _____
2. What is Schering Bridge?
3. Advantages of Schering Bridge?
4. Disadvantages of Schering Bridge?
5. Applications of using Schering bridge are?